Rational fibers of cubic maps and a sharp obstruction to scalar height bounds

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Abstract

A rational function of two rational variables whose numerator and denominator have total degree at most three has an infinite rational fiber in every nonempty real open set. The points on this fiber have unbounded height. The assertion for smooth cubic fibers follows from the tangent map and Mazur's torsion theorem. Thus even a completely ineffective bound on the height of a rational pair in terms of a single rational measurement requires degree at least four, if the bound is to hold on a real open set. This degree threshold is sharp: the rational fibers of $U^4 + V^4$ are all finite, by Faltings' theorem. Both assertions extend to every number field, using Merel's torsion theorem and simultaneous archimedean approximation. These assertions concern ambient rational pairs and do not exclude stronger height bounds on special sets of elliptic curve multiples.

1 The height question

For a rational pair (u, v), write $(1 : u : v) = (c : a : b)$ with coprime integers a, b, c and $c > 0$, and put

$$h_2(u, v) = \log \max\{|a|, |b|, c\}.$$

For $q = a/b \in \mathbb{Q}$ in lowest terms with $b > 0$, let $h(q) = \log \max\{|a|, b\}$. Only finitely many rational pairs have h_2 at most any fixed real number.

Theorem 1.1. *Let $A, B \in \mathbb{Q}[U, V]$ be relatively prime, with $B \neq 0$, $\max\{\deg A, \deg B\} \leq 3$, and A/B nonconstant. For every nonempty open subset $\Omega \subset \mathbb{R}^2$, there exists $c \in \mathbb{Q}$ such that*

$$\{(u, v) \in \Omega \cap \mathbb{Q}^2 : B(u, v) \neq 0,\ A(u, v)/B(u, v) = c\}$$

is infinite. Its h_2-heights are unbounded.

The open set may in particular impose any finite list of strict real inequalities. Thus positivity conditions on two real conjugates do not remove the obstruction in this theorem.

2 The tangent map on a plane cubic

Let $C \subset \mathbb{P}^2$ be a smooth plane cubic over $\mathbb{Q}$ possessing a rational point. Its tangent map $\tau : C \to C$ sends P to the residual point of intersection with the tangent line at P, with intersection multiplicities included. This is a morphism defined over $\mathbb{Q}$. Let H denote the hyperplane divisor class on C.

Lemma 2.1. *For $P \in C(\mathbb{Q})$, put $D_P = H - 3[P] \in \mathrm{Pic}^0(C)(\mathbb{Q})$. If D_P is torsion, then*

$$\tau^{39}(P) = \tau^3(P). \tag{1}$$

The two morphisms τ^{39} and τ^3 are distinct on every smooth plane cubic over a field of characteristic zero.

Proof. Choose an origin $O \in C(\mathbb{Q})$ and identify C with its Jacobian E. Write $H_0 = H - 3[O] \in E(\mathbb{Q})$. The tangent intersection identity $2[P] + [\tau(P)] \sim H$ gives

$$\tau(P) = H_0 - 2P, \qquad \tau^k(P) = (-2)^k P + \frac{1-(-2)^k}{3} H_0.$$

The coefficient in this formula is an integer. Since $D_P = H_0 - 3P$,

$$\tau^{39}(P) - \tau^3(P) = \frac{(-2)^3 - (-2)^{39}}{3} D_P = 183251937960 D_P.$$

Mazur's theorem [1, Theorem (7′)] says that every rational torsion point of an elliptic curve has order in $\{1, 2, \ldots, 10, 12\}$. All these integers divide

$$2520 = \mathrm{lcm}(1, 2, \ldots, 10, 12), \qquad 183251937960 = 2520 \cdot 72719023.$$

This proves (1) for torsion D_P. Finally τ is translation followed by multiplication by -2; hence $\deg \tau^k = 4^k$. The degrees for $k = 39$ and $k = 3$ are unequal, so the morphisms cannot coincide. □

Lemma 2.2. *If $C/\mathbb{Q}$ is a smooth projective genus-one curve, $P \in C(\mathbb{Q})$, and $E = \mathrm{Pic}^0(C)$ has a rational point of infinite order, then every real neighborhood of P in $C(\mathbb{R})$ contains infinitely many rational points of C.*

Proof. Use P as origin and let $D \in E(\mathbb{Q})$ have infinite order. The compact group $E(\mathbb{R})$ contains a sequence of distinct multiples $[m_j]D$ converging to its identity. To see this directly, compactness gives a convergent subsequence of the distinct points $[n]D$; differences of suitable pairs give distinct nonzero multiples converging to the identity. Their translates by P give the required rational points. □

3 Proof of the fiber theorem

Proof of Theorem 1.1. Put $f = A/B$. Delete the zero set of B and the critical locus $df = 0$. Because f is nonconstant in characteristic zero, what remains is a nonempty Zariski open subset $W \subset \mathbb{A}^2$. Every nonempty real open subset of $\mathbb{R}^2$ meets $W(\mathbb{Q})$.

First suppose the projective closure of the generic fiber of f is not a smooth geometrically integral plane cubic. Then there is a nonempty open subset $T \subset \mathbb{A}^1$ such that the fibers over T have this same property and degree at most three. Choose $P \in \Omega \cap W(\mathbb{Q})$ with $c = f(P) \in T(\mathbb{Q})$. The geometric component of $A - cB = 0$ through P is unique because P is a smooth point of the fiber. It is fixed by Galois, and is therefore defined over $\mathbb{Q}$. It is either a line, a conic, or a singular integral plane cubic. Its normalization has genus zero, and the smooth rational point P lifts to a rational point on that normalization. The normalization is consequently isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$. A rational parametrization gives infinitely many rational points arbitrarily close to P; all sufficiently close points belong to Ω and avoid $B = 0$.

Now suppose the generic projective fiber is a smooth geometrically integral plane cubic. After deleting finitely many values of f, the projective fibers are smooth plane cubics.

Tangent intersection in these fibers defines a rational map τ on the total space, defined over $\mathbb{Q}$ and preserving f. Equivalently it defines a rational map of the plane on a nonempty open subset, since the total space of the affine pencil is the graph of f. The rational maps τ^{39} and τ^3 are unequal: their restrictions to a generic smooth cubic fiber have degrees 4^{39} and 4^3, respectively. Thus the locus on which both are defined and different contains a nonempty Zariski open subset of the plane. Choose $P \in \Omega \cap \mathbb{Q}^2$ in this open subset, in W, and above a smooth cubic fiber C. Lemma 2.1 shows that $D_P \in \mathrm{Pic}^0(C)(\mathbb{Q})$ has infinite order. Lemma 2.2 supplies infinitely many rational points of C arbitrarily close to P. A small enough neighborhood stays in the affine chart, in Ω, and away from $B = 0$. This proves the required infinitude in both cases.

The height assertion follows because a bounded-height set of rational pairs is finite. □

Corollary 3.1. *Under the hypotheses of Theorem 1.1, there is no finite-valued function* $\Phi : [0, \infty) \to \mathbb{R}$ *such that*

$$h_2(u, v) \le \Phi\big(h(f(u, v))\big)$$

for every $(u, v) \in \Omega \cap \mathbb{Q}^2$ *with* $B(u, v) \neq 0$. *In particular, no such* f *is finite-to-one on the rational points of any nonempty real open subset of its domain.*

Proof. The infinite fiber in Theorem 1.1 has one fixed value of $h(f(u, v))$ and unbounded $h_2(u, v)$. □

4 Sharpness of the degree threshold

Proposition 4.1. *The polynomial* $g(U, V) = U^4 + V^4$ *is finite-to-one on* $\mathbb{Q}^2$. *Consequently there exists a finite-valued nondecreasing function* $\Phi : [0, \infty) \to [0, \infty)$ *such that*

$$h_2(u, v) \le \Phi\big(h(u^4 + v^4)\big) \qquad (u, v \in \mathbb{Q}).$$

This assertion gives no effective formula for Φ.

Proof. The fiber above zero consists only of $(0, 0)$. For $c \in \mathbb{Q}^\times$, the projective curve

$$U^4 + V^4 = cW^4$$

is smooth over $\overline{\mathbb{Q}}$: its three partial derivatives vanish simultaneously only when $U = V = W = 0$, which is not a projective point. A smooth plane quartic has genus three. Its rational points are finite by Faltings' theorem [2], proving the first assertion. For each real $T \ge 0$ there are only finitely many $c \in \mathbb{Q}$ with $h(c) \le T$. The union of their rational fibers under g is finite. Take $\Phi(T)$ to be the maximum of h_2 over that finite union, including zero in the maximum. This also proves monotonicity. □

Remark 4.2. Finite fibers are substantially weaker than injectivity. Proposition 4.1 does not solve the polynomial-injection problem for $\mathbb{Q}^2 \to \mathbb{Q}$. Poonen's theorem [4] obtains such an injection under a form of the Bombieri–Lang conjecture.

5 Scope for rational height measurements

Suppose a quadratic number field is written $K = \mathbb{Q}(\sqrt{d})$, and a height argument produces an element $S = u + v\sqrt{d}$ with $u, v \in \mathbb{Q}$. If the only restrictions on u, v are strict inequalities in the real embeddings, those restrictions describe a real open set. Corollary 3.1 rules out replacing the two independent rational coordinates by any single rational function with

numerator and denominator of total degree at most three while retaining even an ineffective uniform height bound.

If, however, S is a prescribed sum of abscissas of integer multiples of fixed elliptic points, its attainable coordinate pairs form a much smaller arithmetic set. Theorem 1.1 makes no assertion that its infinite fiber meets this set infinitely often. A height theorem on that special set may therefore hold, and must use the arithmetic restriction rather than positivity alone. Similarly, this result is not a lower bound for the number of quantified variables in a Diophantine definition of the integers.

6 Extension to every number field

Let F be a number field and let

$$F_\infty = \prod_{v|\infty} F_v,$$

where one place is used for each conjugate pair of complex embeddings. We give this product its usual real and complex topology and view F diagonally in it. A $\mathbb{Q}$-basis of F identifies $F \otimes_{\mathbb{Q}} \mathbb{R}$ with F_∞ as a real vector space; the density of $\mathbb{Q}^{[F:\mathbb{Q}]}$ proves that F is dense in F_∞. The same argument shows that the F-points of every nonempty Zariski open subset of $\mathbb{A}^2_F$ meet every nonempty open subset of F_∞^2. For $u, v \in F$ the pair height is the absolute logarithmic projective height $h(1 : u : v)$. Bounded-height subsets of F^2 are finite: for an algebraic number of degree at most $[F : \mathbb{Q}]$ and bounded height, its primitive minimal polynomial has bounded Mahler measure and hence bounded integer coefficients. There are only finitely many such polynomials and their roots. Apply this to u and v, whose heights are each at most $h(1 : u : v)$. This is the form of Northcott's finiteness theorem used below.

Theorem 6.1. *Let $A, B \in F[U, V]$ be relatively prime, with $B \neq 0$, $\max\{\deg A, \deg B\} \leq 3$, and A/B nonconstant. For every nonempty open subset $\Omega \subset F_\infty^2$, there is $c \in F$ such that*

$$\{(u, v) \in F^2 \cap \Omega : B(u, v) \neq 0,\ A(u, v)/B(u, v) = c\}$$

is infinite and has unbounded absolute logarithmic height.

Proof. Merel's uniform torsion theorem [3, Introduction, Corollary] provides an integer $M_F \geq 1$ annihilating the torsion subgroup of $E(F)$ for every elliptic curve E/F. For instance, if N is an integer upper bound for the orders of all such torsion points, one may take $M_F = \operatorname{lcm}(1, \ldots, N)$. Write $3M_F = 2^a u$ with u odd, and choose a positive even integer h such that $2^h \equiv 1 \pmod{u}$. Then

$$M_F \ \mid\ \frac{(-2)^a - (-2)^{a+h}}{3}.$$

On a smooth plane cubic, the tangent-map computation of Lemma 2.1 therefore shows

$$D_P \text{ torsion} \quad \Longrightarrow \quad \tau^{a+h}(P) = \tau^a(P).$$

These maps have distinct degrees. The unequal-iterate locus is thus a nonempty Zariski open set in any pencil with smooth generic cubic fiber. Choose an F-point P in this locus and in the prescribed archimedean open set, avoiding the same finite exceptional set of fibers as in the proof over $\mathbb{Q}$. Then D_P has infinite order.

The group $\prod_{v|\infty} E(F_v)$ is compact. Its infinite cyclic subgroup generated by D_P has a sequence of distinct multiples converging to the identity simultaneously at every archimedean

place. Translation by P gives infinitely many F-points of its fiber inside Ω, avoiding the denominator and the points at infinity after shrinking the neighborhoods.

If the generic projective fiber is not a smooth geometrically integral cubic, the component through a smooth F-point is a rational curve, exactly as in the proof of Theorem 1.1. Its normalization is $\mathbb{P}^1_F$, and density of F in F_∞ gives infinitely many points simultaneously close to the chosen point. Finally, bounded-height subsets of F^2 are finite by Northcott's finiteness theorem, so the heights in the infinite fiber are unbounded. □

Corollary 6.2. *Over every number field F, degree four is the least possible degree of a rational scalar measurement of two variables that has finite fibers on all of F^2. More precisely, no rational function of numerator and denominator degrees at most three is finite-to-one on the F-points of any nonempty archimedean open set, whereas for every $\delta \in F^\times \setminus F^{\times 4}$ the polynomial*

$$g_\delta(U, V) = U^4 - \delta V^4$$

is finite-to-one on F^2. There is a finite-valued nondecreasing function Φ_F such that $h(1 : u : v) \le \Phi_F(h(g_\delta(u, v)))$ for all $u, v \in F$.

Proof. The lower bound follows from Theorem 6.1. The zero fiber of g_δ contains only $(0, 0)$: a point with $v \neq 0$ would make $\delta = (u/v)^4$. For $c \neq 0$, the projective curve $U^4 - \delta V^4 = cW^4$ is smooth of genus three. Its F-points are finite by Faltings' theorem. Northcott's theorem now gives the height function Φ_F by the same finite-union argument as in Proposition 4.1. Such a δ exists in every number field: any rational prime unramified in F has valuation one at every prime of F above it, and hence cannot be a fourth power in F. □

Neither the proof of Theorem 6.1 nor the sharpness argument assigns a numerical height bound. The assertions remain separate from the problem of compressing a prescribed arithmetic subset of elliptic multiples.

References

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